TravelTide Rewards Program: Detailed Report

Introduction, Project Context & Analytical Approach

1. Introduction & Project Context

This report documents the full project lifecycle and process behind TravelTide's customer rewards perk assignments. It is intended for business stakeholders who want to understand the findings in depth, and technical reviewers who need to evaluate the methodology, data engineering decisions, and validation rigor.
The project was completed as part of the Masterschool Data Science Foundations Mastery Program.

The primary business objective was to assign each cohort users exactly one of five reward perks:
Exclusive Discounts, No Change Fees, Free Checked Bags, Free Hotel Meals, and Premium Travel Benefits - based on demonstrated business relevant patterns.
A secondary objective was to generate and document actionable business insights at both session-level and user-level aggregations.

2. Pipeline Architecture: V1 to V2

The project went through two complete pipeline versions. Version 1 reached clustering and initial perk assignment before multiple data quality issues were discovered that required architectural rebuilds rather than patches. The decision to rebuild from scratch rather than patch V1 was deliberate because patching would have produced unreliable feature values with unclear validity.

V1 Issues Requiring V2 Rebuild

- Discount rates >100% from session-level double-counting >> required trip-level deduplication CTE
- 1,336 pre-cohort sessions incorrectly included >> missing date filter on cohort JOIN
- session_duration_sec only populated for flight sessions >> fixed via session_end minus session_start
- booking_lead_time_days excluded hotel-only bookings >> fixed via LEAST(departure_time, check_in_time)
- 7 orphaned cancellation sessions with no matching booking in cohort window >> resolved via pre-cutoff booking inserts as documented exceptions
- 4 deprecated IATA codes (JRS, SXF, THF, TXL) causing NULL passport_required values on valid pre-cohort rows

3. Technical Environment

Database: PostgreSQL (local, tt_project schema) - source tables untouched for referential integrity; all analytical tables are derived copies in the tt_project schema.

IDE: VS Code with Jupyter Notebooks & SQL files. *.ipynb files used for data analysis and ML steps, *.sql files for DB engineering steps. This hybrid approach deliberately separates DB engineer tasks from data science tasks and analysis.

Python: SQLAlchemy for DB reads/writes, pandas for manipulation, scikit-learn for clustering (MinMaxScaler, StandardScaler, KMeans), PyTorch/CUDA GPU environment (NVIDIA RTX 2060), see requirements.txt.

Compatibility: NumPy/PyTorch version conflict resolved after updating all environment versions and rerunning. See requirements.txt.

4. Data Engineering: PostgreSQL Architecture

The V2 data warehouse uses a deliberate duplication strategy: source tables (users, sessions, flights, hotels) remain untouched as the source of truth. All analytical tables are derived and stored separately in the tt_project schema. This design strategy creates a repeatable pipeline process of engineered data replication, without risk of source data corruption.

cohort23	522 cohort user IDs fit the eligibility filter = 7+ sessions since Jan 4, 2023.
cohort_analytics	49,218 rows × 60 fields. Primary analytical data lake. One row per session, cohort users only. Session type, booking type, passport flag, haversine distance, and all engineered session fields added here.
session_agg	One row per user. Website usage aggregations incl: sessions, bookings, browse rate, conversion rate, duration, clicks.
user_profile	One row per user. Demographics and account metadata incl: birthdate, gender, married, has_children, home_airport, sign_up_date, first/last session dates.
trip_agg	One row per user. Completed trip financial and distance metrics incl: gross/net spend, fare per seat, hotel rates, nights, lead time, package/hotel/flight split, distance aggregations.
trip_cx_agg	One row per user. Cancelled trip financial metrics: identical structure to trip_agg but filtered to cancelled trips. Enables independent completed vs cancelled spend analysis.
user_loyalty	One row per user. Lifetime loyalty metrics incl: life_total_trips (pre-cohort + cohort), cancellation counts, days_since_signup, first_purchase_date.
user_features	One row per user. 109-column FINAL feature table joining all five intermediate tables. Input to all ML and perk assignment processes. Complete aggregate table prior to ML transformations.
user_features_transformed	user_features + dtype fixes + zero fills + outlier caps + log transforms + behavioral bins + cluster labels + segment labels. ML-ready checkpoint with 160-columns.
perk_assignments	5,998 rows × 4 columns: user_id, assigned_perk, life_total_trips, assignment_reason. Final deliverable table.

5. Key Feature Engineering Decisions

session_type field: Direct result of V1 lessons. “browse” / “booking” / “cancel” classification added to cohort_analytics. Critical for separating engagement metrics from transactional metrics. Cancel sessions excluded from financial aggregations entirely.

cancelled_all segment: 88 users who booked but cancelled 100% of their trips treated as categorically distinct from never-bookers. never_booked = never attempted; cancelled_all = attempted and reversed. Assigned No Change Fees, not Exclusive Discounts.

Haversine distance: Calculated in Python via SQLAlchemy parameterized UPDATE on session_id. 14,277 rows populated. Distributed across session_agg (attempted), trip_agg (completed), and trip_cx_agg (cancelled).

passport_required boolean: Intl trip flag derived from home_airport vs destination_airport country code comparison. Four deprecated IATA codes (JRS, SXF, THF, TXL) added as retroactive fix to NULL values in airport_lookup reference table on 6 pre-cohort exception rows.

COALESCE architecture: All LEFT JOIN fields wrapped in COALESCE to prevent NULL propagation in downstream aggregations. NULL-safe division via NULLIF(denominator, 0) pattern applied consistently across all ratio calculations. Integer division resolved via ::float casts.

booking_lead_time_days: LEAST(departure_time, check_in_time) used as the booking reference date to handle hotel- & flight-only bookings. Direct result of V1 using only departure_time, producing NULL lead times for ~2,400 hotel-only sessions.

6. Data Preparation for ML

Prior to clustering, user_features was transformed through four sequential steps applied only to the ML input, NOT the base user_features table, preserving interpretable original values for validation and business insight queries.

Step 1a. Dtype fixes & zero-fills: Date columns cast to datetime. Numeric fields with meaningful zeros (cancel rates, discount rates, booking counts for non-bookers) filled with 0 rather than excluded. 22 users adjusted to include the pre-cohort booking half of a cancel row.

Step 1b. Outlier capping: Three-chart diagnostic approach: histogram + boxplot + log1p histogram per field. Zero-inflated fields evaluated on non-zero populations only. Gross and net spend fields capped independently. 10 fields capped at 95th or 99th percentile.

Step 1c. Log transforms: 24 log_ prefixed columns created via log1p transform. Applied to highly skewed continuous fields (base_fare_usd skew 8.56, hotel_per_room_usd skew 2.35). Original fields retained alongside transforms.

Step 1d. Behavioral bins: Session duration bins (quick/moderate/deep) and page click bins dropped after confirming unimodal user-level distributions. Binning session-level patterns into user-level aggregations removed signal rather than adding. Booking lead time bins revised from V1 session-level boundaries to four user-level groups.

7. PCA & Two-Path Clustering Architecture

PCA was run twice. The first iteration used a manually curated 35-feature set — this was rejected because manual curation introduced analyst bias before the model had a chance to find its own structure. The second iteration used the full clean 64-feature set, revealing eight meaningful components explaining 67% of total variance. This informed a two-path parallel clustering design to compare dimension groupings. Path A 'standard' grouped business features, Path B - PCA (consolidated).

Dimension	Path	k Selected	Silhouette	Notes
Website (A)	A	2	0.38	Active bookers vs browsers/low-converters
Spending (A)	A	3	0.41	Standard spenders, non-spenders, hotel-only domestic
Travel (A)	A	3	0.44	Standard roundtrip, non-travelers, hotel-centric short-haul
Cancellation (A)	A	5*	0.31*	*Zero-inflation correction applied — see below
Spending+Distance (B)	B	3	0.42	Mainstream, non-spenders, hotel-only domestic
Cancellation (B)	B	5*	0.31*	Identical to Path A after correction
Hotel Behavior (B)	B	3	0.39	Flight-led, regular hotel bookers, luxury hotel discount
Flight Discounts (B)	B	3	0.45	Non-discount, heavy discount, high-volume package
Intl Travel (B)	B	4	0.48	Domestic only, frequent mixed, dedicated intl, intl cancellers

8. Zero-Inflation Problem & Cancellation Fix

The cancellation dimension initially returned a silhouette score of 0.955 — which was identified as artificially inflated by outlier cap null-filll. 90.1% of users have zero cancellation activity. K-Means was finding the zero-cluster vs non-zero-cluster split, not genuine behavioral subgroups. Resolution: cluster only the 595 actual cancellers, assign the 5,403 non-cancellers to a 'non_cancellers' cluster, cancellers-only confirmed k=5 as best choice (silhouette 0.31 - Nominal, but genuine signal).

9. Segment Naming & Critical Limitation

Cluster segment names were assigned post-hoc by profiling cluster centroids after K-Means fitted. Initial naming based on 'stand out' parts of the mean profile. Validation showed several clusters somewhat misleading – notably 'flight_advanced_planner_cancellers' (avg lead time = 7 days, below baseline) and 'highvolume_package_travelers' (actually a mixed signal of flight discount rate AND package volume). These mixed 'signals' were the primary reason strict ML-based perk assignment was abandoned in favor of independent rule-based logic, allowing application of signal insights from geometrically cluster separation, without dependency on ML processes for perk assignment.

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Perk Assignment Logic, Cross-Validation & Business Insights

10. Perk Assignment: using Lifecycle-Gated Rule-Based System

The final perk assignment system was built independently of the ML clustering output, using direct SQL threshold logic applied to user_features as a deliberate validation strategy. If independent rule-based logic converged on the same user populations as ML clustering, that convergence would validate both 'path' approaches simultaneously. The convergence confirmed as: "dedicated_intl_travelers" cluster (931 users) = rule-based ≥90% intl threshold (959 users); "luxury_hotel discount_seekers" (290 users) = hotel_discount_rate ≥50% threshold (297 users). The populations confirmed the ML findings and applied real behavioral boundary logic.

The assignment uses four lifecycle tiers applied sequentially:

Tier 0 (life_total_trips = 0): 522 users --> Exclusive Discounts unconditionally. No behavioral signal exists; acquisition hook only. Never-booked users average 570 days on platform ~ same window as all other tiers.

Tier 1 (life_total_trips = 1): 939 users --> cancel_spend_ratio ≥ 0.40 == No Change Fees (100 users). All others == Exclusive Discounts (839 users). International travel percentage statistically unreliable at n=1 --> excluded from Tier 1 logic.

Tier 2 (life_total_trips = 2): 1,315 users --> COALESCE(cancel_spend_ratio,0) ≥ 0.40== No Change Fees (77 users). COALESCE(pct_complete_flights_intl,0) = 1.0 AND cancel <0.40== Free Checked Bags (302 users). Remaining: Exclusive Discounts (936 users). NULL handling was critical since 1,173 users had NULL cancel_spend_ratio (zero cancellation activity).

Tier 3+ (life_total_trips ≥ 3): 3,222 users --> Full behavioral evaluation in priority order: (1) cancel_spend_ratio ≥ 0.40 == No Change Fees [56]; (2) pct_complete_flights_intl ≥ 0.90 == Free Checked Bags [413]; (3) hotel_discount_rate ≥ 0.50 AND flight_discount_rate <0.50 == Free Hotel Meals [221]; (4) flight_discount_rate ≥ 0.50 AND hotel_discount_rate <0.50 == Exclusive Discounts [249]; (5) residual --> Premium Travel Benefits [2,283].

11. Threshold Decisions & Forward-Looking Notes

pct_complete_flights_intl ≥ 0.90: Natural data gap confirmed: zero users exist between 0.90 and 0.95. Threshold is a behavioral boundary, not arbitrary. 34 additional users exist at 0.80–0.89 (majority-international but with meaningful domestic mix). Revisit trigger: cohort growth or checked bag cost data.

cancel_spend_ratio ≥ 0.40: Captures all 88 cancelled_all users (ratio = 1.0) plus high behavioral cancellers. The 0.30 threshold adds 39 users. The 0.40 threshold selected because avg cancel_spend_ratio for captured group = 0.662, confirming genuine high-cancel users, not borderline cases.

hotel_discount_rate ≥ 0.50: Mirrors luxury_hotel_discount_seekers cluster (290 users) almost exactly (297 at threshold). Dual-discount exclusion (flight_discount_rate <0.50) prevents price-sensitive users from landing in Free Hotel Meals rather than Exclusive Discounts.

life_total_trips ≥ 3 for full evaluation: Statistical basis: 1 point = dot, 2 points = line segment, 3 points = trend. Single-trip users show avg cancel_ratio 0.61 (noise from single-event data). By three trips, cancel ratio stabilizes at 0.31. Full behavioral pattern only becomes reliable after 3+ observations.

12. Additional Business Insights: Unique to This Report

25-34 cohort: the strategic inflection point

51.4% of 25-34 users are assigned Exclusive Discounts, 27.6% Premium Travel Benefits, the most balanced of any age band. This age group is actively transitioning between lifecycle tiers (including age-based travel restrictions ex. rental car limitations). Targeted conversion incentives for this band have the highest probability of creating long-term loyal customers.

65+ behavioral paradox

65+ users pay the highest avg fare per seat ($563, 51% above cohort avg) and take the longest trips (6.68 avg nights) but have the lowest booking frequency (2.64 trips/year) and high cancellation rates (69%). These are premium occasional travelers, not loyal regulars. Perks designed for frequency (Premium Travel Benefits) are inappropriate for this group, which is correctly dominated by Exclusive Discounts (84.9%).

Flight-only cancellation risk

Flight-only bookings represent 11.1% of all bookings but 31.6% of all cancellation sessions, 3x disproportionate cancellation risk. Combined with extreme lead time skew (44.9 days avg vs 7 days median), this pattern is consistent with speculative advance booking: users reserving flights with long lead times and cancelling when plans change.

Package progression signal

Hotel-only bookings represent 31.8% of Tier 1 trips but only 9.2% of Tier 4 trips, a 22-point drop as users progress through the lifecycle. Package (flight+hotel) booking rate increases from 64.9% at Tier 1 to 73.8% at Tier 4. Package booking rate is a measurable proxy for lifecycle progression and in addition to `life_total_trips` could serve as loyalty program milestone triggers.